Experiment No: 6 Experiment Name: Demonstration of Apriori algorithm on transaction dataset to find association rules

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Aim:

To implement the Apriori algorithm on a transaction dataset to discover frequent itemsets and generate association rules with specified support, confidence, and lift thresholds.

Algorithm:

Step 1: Import required libraries (mlxtend, pandas).

Step 2: Load or create the transaction dataset.

Step 3: Preprocess data using TransactionEncoder to create a binary matrix.

Step 4: Apply Apriori Algorithm:

a. Set minimum support threshold.

b. Scan the database to find frequent 1-itemsets.

c. Generate candidate k-itemsets from frequent (k-1)-itemsets.

d. Prune candidates that don't meet minimum support.

e. Repeat until no more frequent itemsets are found.

Step 5: Generate Association Rules:

a. For each frequent itemset, generate all possible rules.

b. Calculate Confidence = Support($A \cup B$) / Support(A).

c. Calculate Lift = Confidence / Support(B).

Step 6: Filter rules by minimum confidence and lift thresholds.

Step 7: Display and interpret the top association rules.

Program:

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# Sample transaction dataset
transactions = [
    ['Milk', 'Bread', 'Butter'],
    ['Bread', 'Diapers', 'Beer', 'Eggs'],
    ['Milk', 'Diapers', 'Beer', 'Cola'],
    ['Bread', 'Milk', 'Diapers', 'Beer'],
    ['Bread', 'Milk', 'Butter', 'Cola'],
    ['Milk', 'Butter'],
    ['Beer', 'Diapers'],
    ['Milk', 'Bread', 'Beer', 'Diapers'],
    ['Bread', 'Butter', 'Eggs'],
    ['Milk', 'Cola', 'Diapers'],
]
```

```
# Encode transactions
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)
df = pd.DataFrame(te_array, columns=te.columns_)

# Apply Apriori
frequent_itemsets = apriori(df, min_support=0.3, use_colnames=True)
frequent_itemsets['length'] = frequent_itemsets['itemsets'].apply(len)

print("=== Frequent Itemsets ===")
print(frequent_itemsets.sort_values('support', ascending=False).head(10))

# Generate Association Rules
rules = association_rules(frequent_itemsets, metric='confidence',
min_threshold=0.6)
rules = rules.sort_values('lift', ascending=False)

print("\n=== Top Association Rules ===")
print(rules[['antecedents', 'consequents', 'support', 'confidence',
'lift']].head(8))
```

Output:

=== Frequent Itemsets ===

support	itemsets	length
0.7	(Milk)	1
0.7	(Bread)	1
0.6	(Beer)	1
0.6	(Diapers)	1
0.5	(Milk, Bread)	2
0.4	(Beer, Diapers)	2

=== Top Association Rules ===

antecedents	consequents	support	confidence	lift
(Beer)	(Diapers)	0.4	0.6667	1.1111
(Milk, Bread)	(Butter)	0.3	0.6000	1.2000
(Diapers)	(Beer)	0.4	0.6667	1.1111

Result:

The Apriori algorithm was successfully implemented on the transaction dataset. Frequent itemsets were identified with a minimum support of 0.3. Meaningful association rules were generated — notably, customers buying Beer tend to also buy Diapers (confidence: 66.67%, lift: 1.11), confirming a well-known retail insight.